



Cheyenne Regional Airport

City Council Work Session - Master Plan Overview

December 12, 2025



Agenda



Master Plan Overview



Project Schedule & Milestones



Aviation Demand Forecasts



Facility Requirements



Land Use Analysis



Next Steps

What is an Airport Master Plan?

“An airport master plan is a comprehensive study of an airport and usually describes the short-, medium-, and long-term development plans to meet future aviation demand.”

-FAA Advisory Circular 150/5070-6B Airport Master Plans

How often is an Airport Master Plan updated?

Updated roughly every 10 years, depending on the airport's growth, changes in aviation demand, and regulatory requirements.

-Last CYS master plan was completed in 2014

What is the purpose of an Airport Master Plan?

+ A plan that . . .

- Provides guidance for future development
- Satisfies future aviation demand
- Is financially feasible
- Identifies and responds to environmental and socioeconomic conditions



What are the goals of an Airport Master Plan?

Prepare the airport for the future by thoroughly understanding its **infrastructure requirements**.

Maintain/enhance **safe operations** at the Airport.

Foster stakeholder engagement through **open communication**.

Develop a plan to **optimize airport land** by maximizing the use of available space.

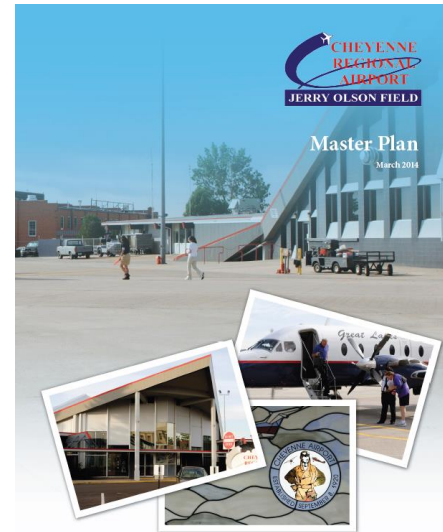
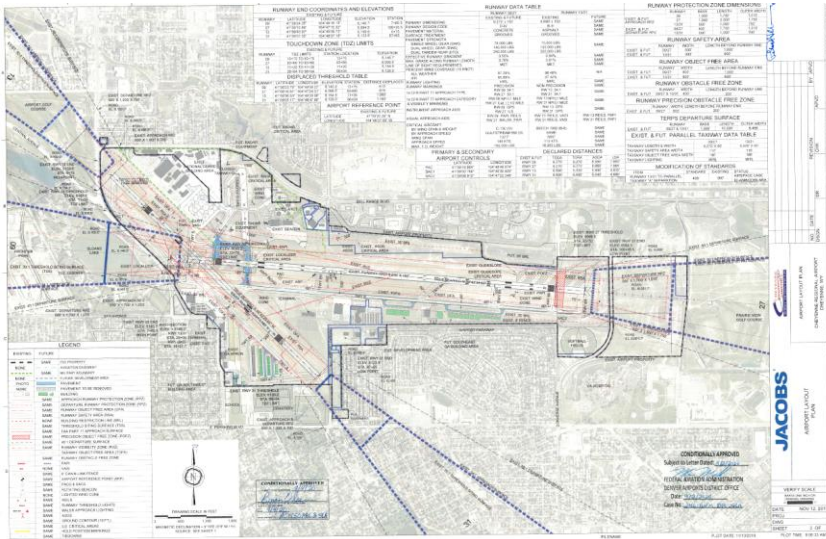
Enhance the **efficiency of operations** of the Airport.

Development program that promotes **fiscal sustainability**.

What do you get from an Airport Master Plan?

*Airport Layout Plan
(illustrates the plan)*

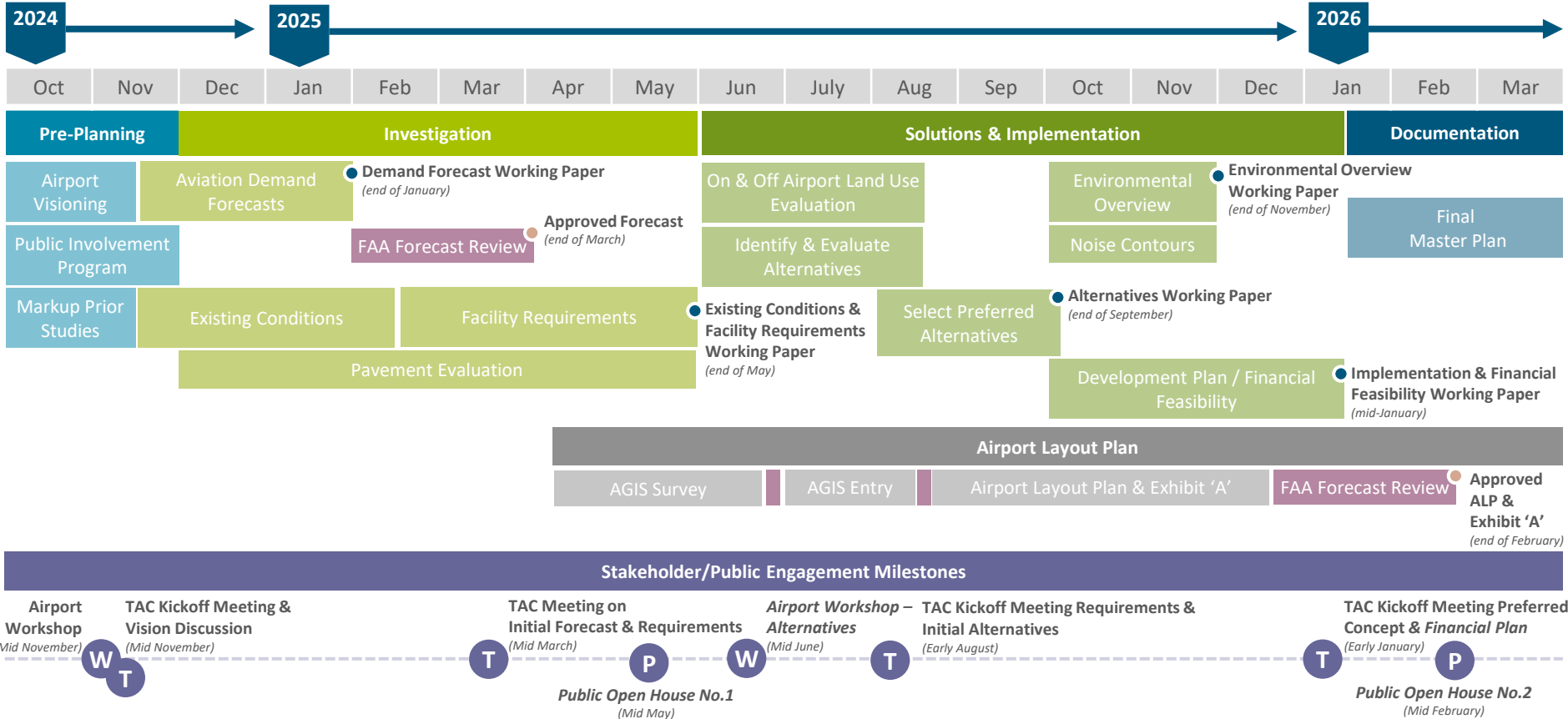
*Working Papers/Final Report
(documents the why and how)*



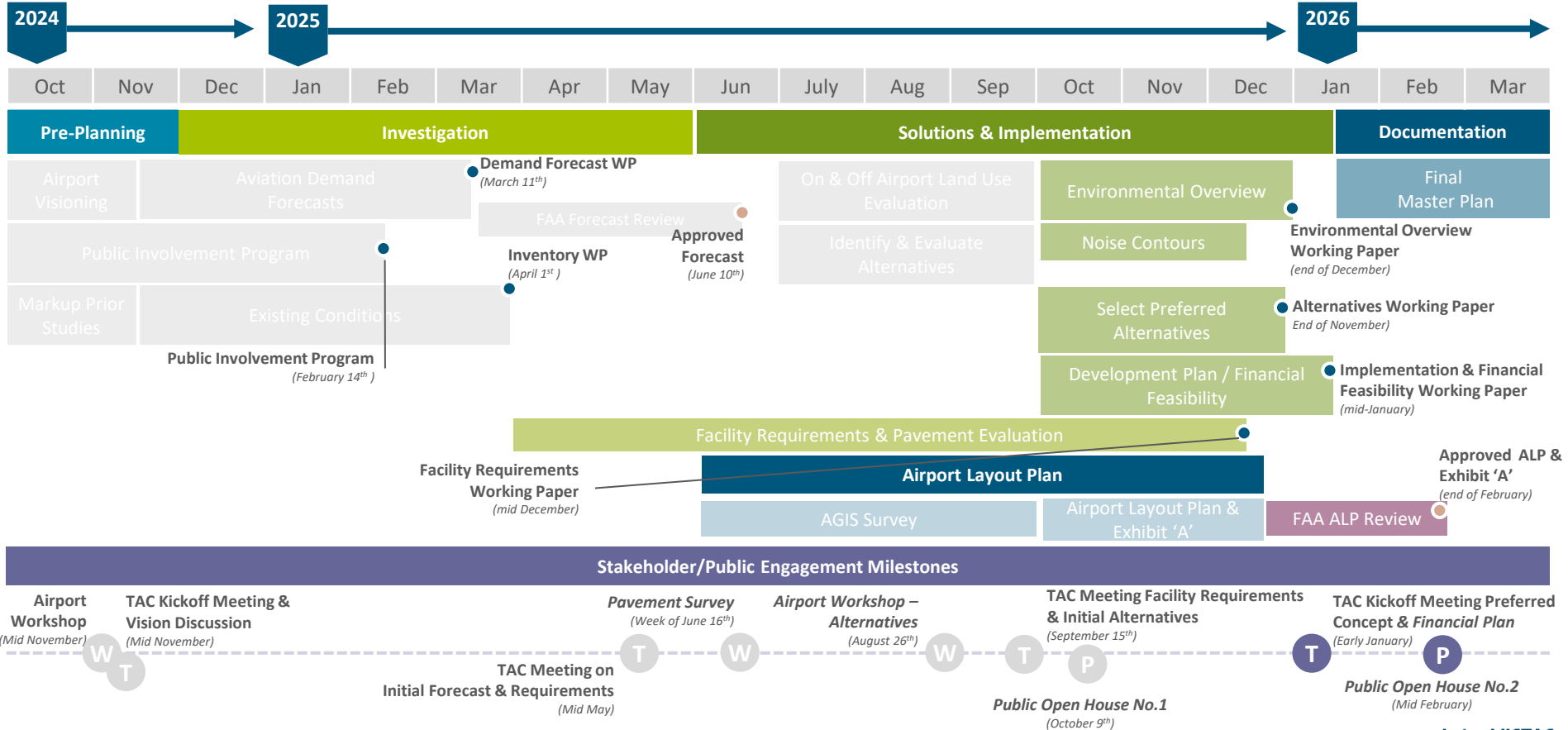
Master Plan Scope of Work



Airport Master Plan Schedule



Airport Master Plan Schedule





Aviation Demand Forecasts

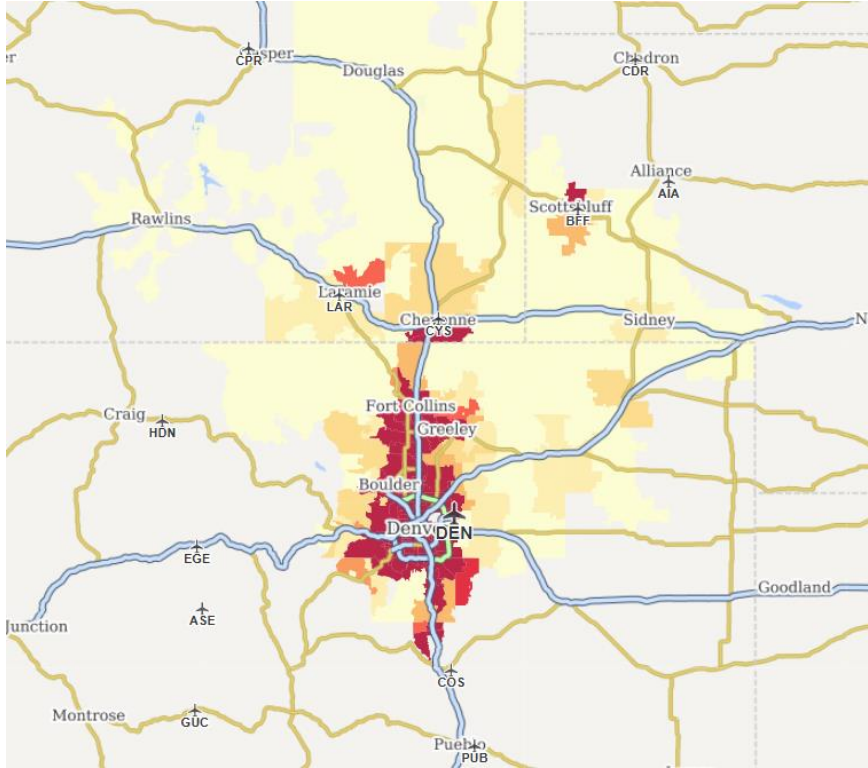


Aviation Demand Forecasts Overview

- + **Task Goal:** Establish accurate baseline data and forecast aviation activity such as enplanements, operations and based aircraft
- + **Task Components:**
 - Collect historical traffic data and socioeconomic activity
 - Analyze trends
 - Develop forecast scenarios
 - Compare against the FAA Terminal Area Forecast
 - Identify a critical aircraft



Airport Service Area



City/Airport	Distance (miles)	2024 Enplaned Passengers
Cheyenne, WY (CYS)	--	27,808
Fort Collins, CO	52	3,052
Laramie, WY (LAR)	54	19,053
Denver, CO (DEN)	108	40,183,334
Scottsbluff, NE (BFF)	115	15,459

Population density, within 150 miles driving distance from CYS

Source: Cirium Diio Mi, Jan 2025

Socio-Economic Activity

	United States	Rocky Mountain Region*	Wyoming State	Cheyenne MSA
Historical CAGR 2003-2023				
Population	0.7%	1.5%	0.7%	0.9%
Employment	1.3%	2.1%	1.4%	1.9%
GRP	2.2%	3.1%	2.3%	2.7%
Per Capita Income	1.7%	2.1%	2.0%	0.9%
Projected CAGR 2023-2043				
Population	0.6%	1.0%	0.6%	0.8%
Employment	1.1%	1.4%	0.8%	1.0%
GRP	1.9%	2.1%	1.8%	1.9%
Per Capita Income	1.5%	1.4%	1.2%	1.6%

Source: Woods and Poole, 2024

Note:* Includes Colorado, Idaho, Montana, Utah, and Wyoming

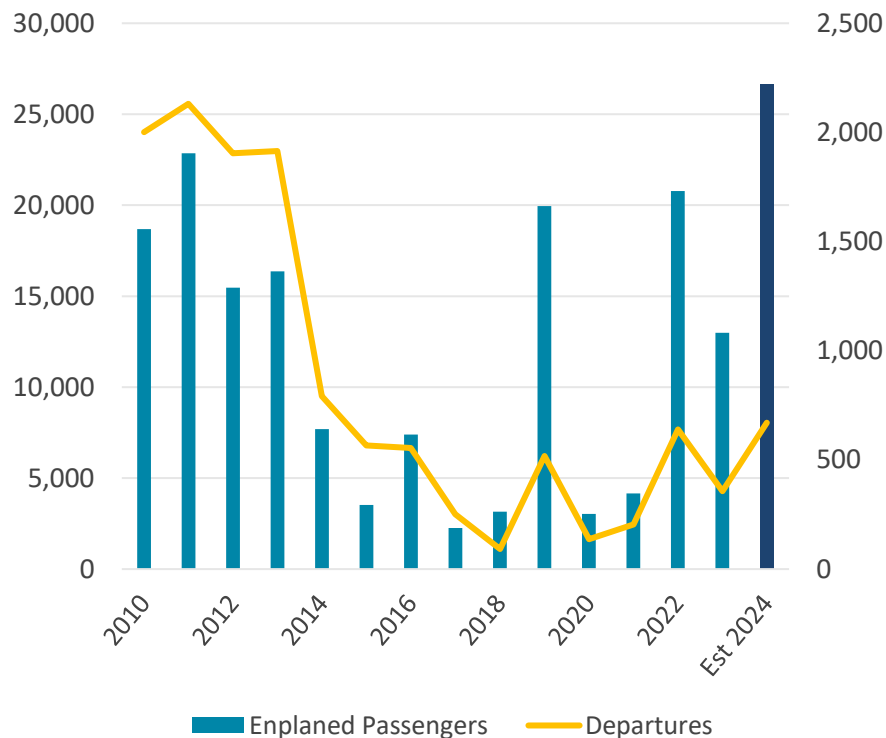
Historical Growth (2003-2023):

- Cheyenne MSA has outpaced Wyoming and U.S. in **employment** and **GRP** growth.
- **Per capita income** growth has lagged behind the regional and national averages.

Projected Growth (2023-2043):

- Cheyenne MSA is expected to experience moderate **population** growth and maintain **employment** and **GRP** growth at or above the national average.
- **Per capita income** growth is projected to be somewhat higher than national and regional averages in the future.

Historical Enplanements



Source: USDOT T100 data, Dec 2024

2010-2011 Peak: American Airlines and Great Lakes Aviation services led to a peak of over 20,000 annual enplaned passengers.

Post-2011 Decline: Termination of American Airlines' service and reduced flights from Great Lakes.

2019 Rebound: American Eagle launch of direct flights to Dallas/Fort Worth led to a significant increase in passengers.

Pandemic Impact: COVID-19 and airport closures temporarily disrupted services.

2021 Recovery: SkyWest resumed daily Denver flights, beginning recovery of passengers.

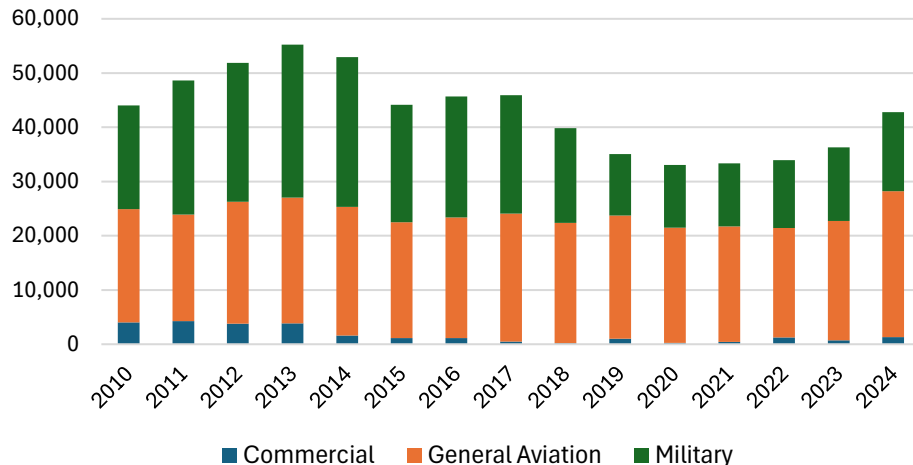
2022 Recovery: Enplaned passengers recovered back to historical peak levels.

2023 Decline: Runway renovations limited flight operations, leading to a decrease in passengers.

2024 Projection: Recovery and growth, with passenger numbers projected to reach 26,600 in 2024.

Historical Aircraft Operations

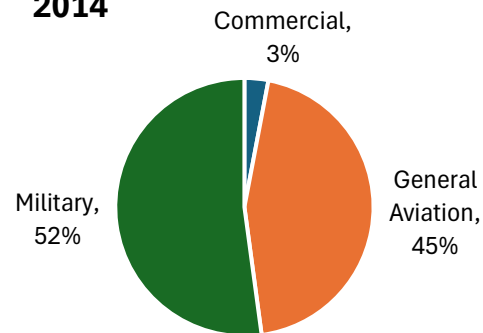
Historical Aircraft Operations



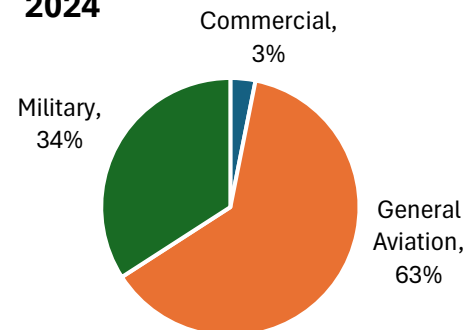
- Decline from peak of 2013, but recent increases after 2020
- Most of the historical change is due to decline in Military operations
- GA is now 63% of total, compared to 45% 10 years earlier
- Need to confirm data for 2023 and 2024

Source: USDOT T100 and FAA ATADS data, Dec 2024

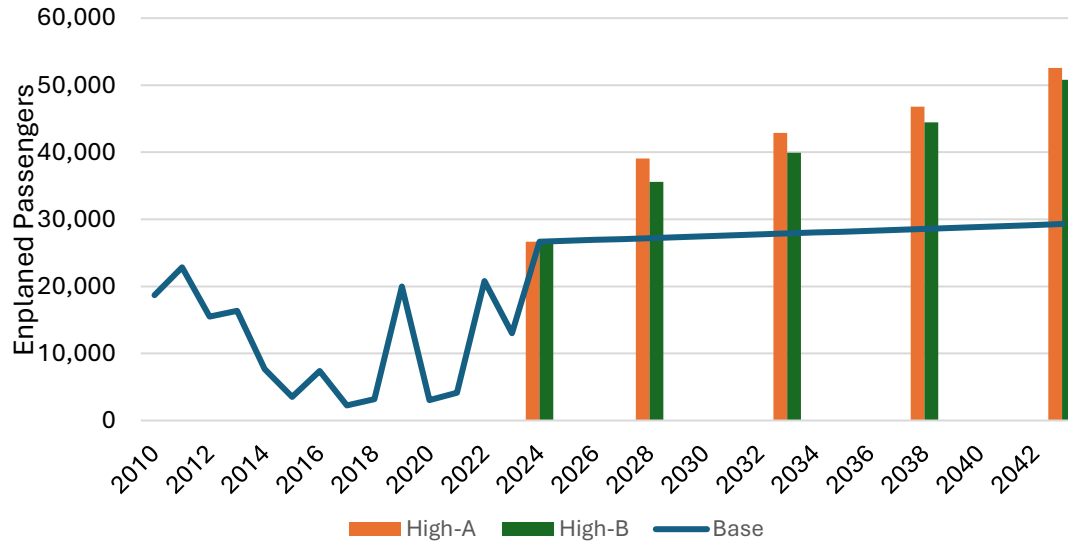
2014



2024



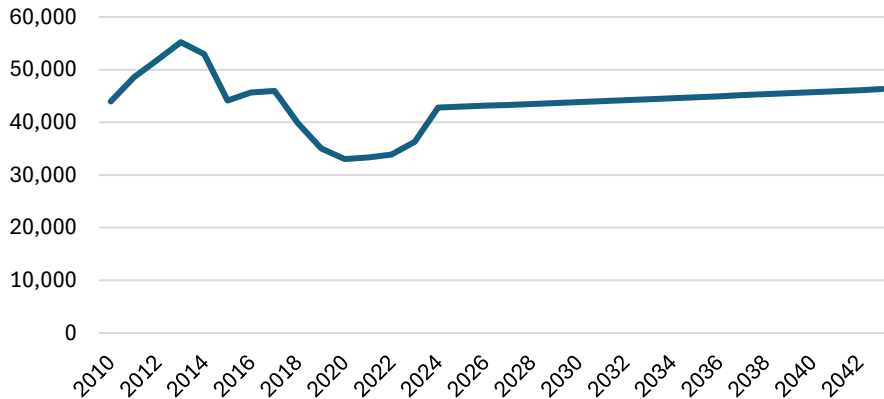
Forecast Enplanements



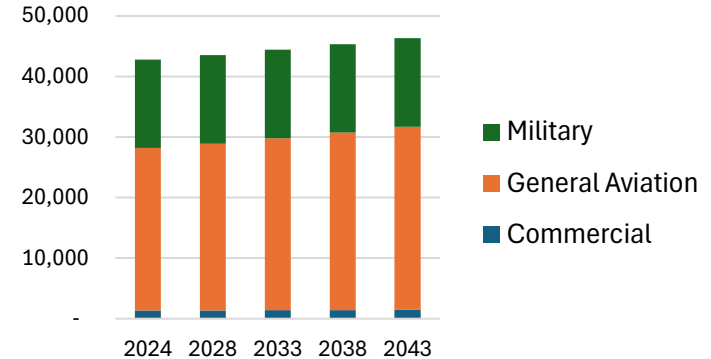
- **Base Case:** current level of flight service to the Denver (DEN) hub will continue, with a conservative 0.5% annual growth rate in passenger numbers
- **Scenario A - Service Expansion to a Second Hub:** CYS will begin service to a second hub starting in 2028
- **Scenario B - Increased Frequency and Larger Aircraft:** the airport will continue service to a single hub (DEN), but with larger aircraft and an increased flight frequency starting in 2028

Forecast Aircraft Operations

Forecast Aircraft Operations



Forecast Operations by Category



- Modest increase in total aircraft operations, 0.4% per year, to 46,000 annual operations in 2043, primarily from GA operations
- GA operations increase about 0.6% per year, a slight premium to FAA's national average growth rate
- Low levels of commercial operations
- No assumed growth in military operations

Forecast Based Aircraft

56

Existing number of based
aircraft

64

Projected number of based
aircraft by 2043

Existing & Future Critical Aircraft





Facility Requirements



Task 05: Facility Requirements

Task Goal: Assess the ability of existing facilities at CYS to accommodate the aviation demand activity forecasts and identify when facility additions and improvements may be needed

Key Subtasks:

- + Identify Planning Activity Levels (PALs)
- + Industry & Emerging Trends Analysis
- + Determine Appropriate Level of Service
- + Facility Requirements Analysis



Planning Activity Levels

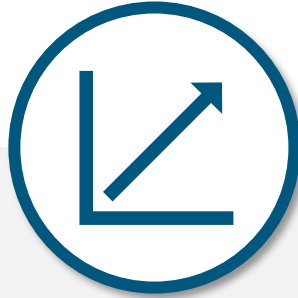
- + **Managing Uncertainty:** Planning activity levels (PALs) provide a flexible approach to addressing the inherent uncertainty in long-range demand forecasting, allowing airports to adapt to actual activity levels rather than fixed timelines
- + **Trigger-Based Development:** PALs help link facility expansions, such as terminal or fuel farm improvements, to specific demand drivers like passenger growth or increased aircraft operations, ensuring projects are implemented as needed
- + **Strategic Capital Planning:** By disassociating projects from specific years, PALs enable airport management to adjust capital investments dynamically, accelerating or decelerating projects based on real-world demand trends

Activity	Base Year (2024)	PAL 1 (Likely +5 years)	PAL 2 (Likely +10 years)	PAL 3 (Likely +20 years)	PAL 4 (Beyond 20 years)
Aircraft operations	42,803	43,500	44,400	46,300	48,000
Based aircraft	58	60	61	64	68
Enplanements	27,808	28,400	29,100	30,600	32,000

Facility Requirement Pillars



Emerging Trends
in Aviation



Ability to
Accommodate
Projected
Demand



Regulatory
Compliance



Enhancing
Airport Services
&
Competitiveness

Emerging Trends in Aviation

Next Gen Airspace

FAA's NextGEN initiative aims to modernize the National Airspace System (NAS) through satellite-enabled navigation, digital communication, and automation

Sustainable Pavement Materials

- Warm Mix Asphalt (WMA)
- Reclaimed Asphalt Pavement (RAP)
- Recycled Aggregate

Terminal Technology

- Security biometric technology to reduce wait times
- Faster passenger and baggage claim processing
- Flexible and multi-purpose use of airport space

TNC Evolution

- Transportation Network Companies (TNC), such as Uber and Lyft, have stabilized after rapid growth
- It remains to be seen what long-term effect this will have on personal vehicle ownership, parking and rental car use

Bus "Flights"

This involves passengers checking bags and going through security at their local airport and being bused to larger hub airports

Declining Demand in Small GA

- Recent FAA Surveys indicate a decline in the number of active general aviation aircraft and flight hours
- Contributing factors include increased fuel prices and training costs

Sustainable Aviation Fuel

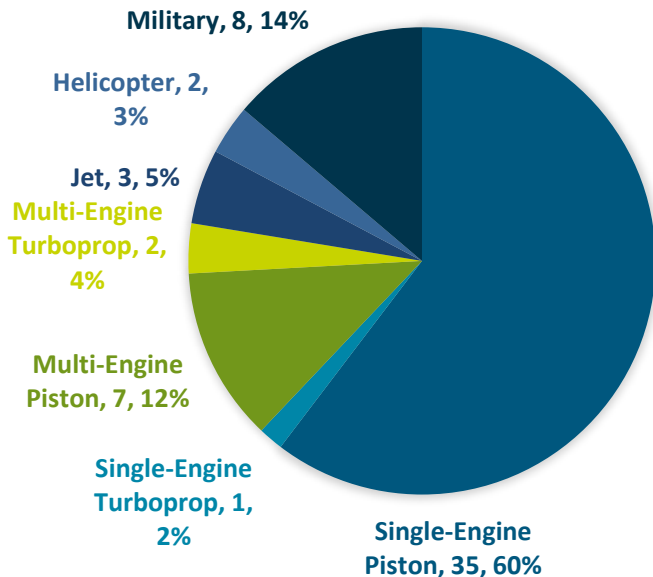
FAA is committed to achieving net-zero greenhouse gas emissions from the U.S. aviation sector by 2050

Multi Functional Equipment

Vehicles that can perform a variety of maintenance tasks by swapping attachments (mowing grass, snowplowing, sweeping, etc.)

Airfield Capacity Analysis

BASED AIRCRAFT, 2024



- As per FAA AC 150/5060-5, Air Service Volume (ASV) is the estimated capacity of the airfield in terms of operations per year, used for long-term planning
- CYS ASV = 215,000 operations/year
- Based on the existing breakdown of based aircraft and operations, it was assumed that 50% percent of operations at CYS are between 12,500 lbs - 300,000 lbs weight class for each year
- No additional runway is needed in the planning period**

	Existing	Forecasted			
	Base Year (2024)	PAL 1 (Likely +5 years)	PAL 2 (Likely +10 years)	PAL 3 (Likely +20 years)	PAL 4 (Beyond 20 years)
Total A/C Operations	42,803	43,500	44,400	46,300	48,000
Existing ASV	215,000	215,000	215,000	215,000	215,000
ASV Ratio	19.9%	20.2%	20.7%	21.5%	22.3%

InterVISTAS, April 2025

Runway Length & Range Planning Parameters

- + To determine if additional runway length is required to accommodate the aircraft shown in the graphic on the right, planning parameters and assumptions were established.
- + Planning parameters and assumptions:
 - Dry pavement conditions
 - Attributes specific to CYS, such as reference elevation and runway gradient
 - Zero wind speed
 - Temperature based on International Standard Atmosphere (ISA) + 15°C.
 - Aircraft analyzed in this evaluation are illustrated in the figure on the right
 - Aircraft planning manuals were used to determine aircraft range
 - Maximum fuel payload, if possible
 - Assumed existing pavement can accommodate evaluated aircraft

Aircraft Evaluated at MTOW



King Air 350 (Rwy 13-31)



Beechcraft 1900 (Rwy 13-31)



C-130 (Rwy 09-27)



EMB-135 (Rwy 09-27)



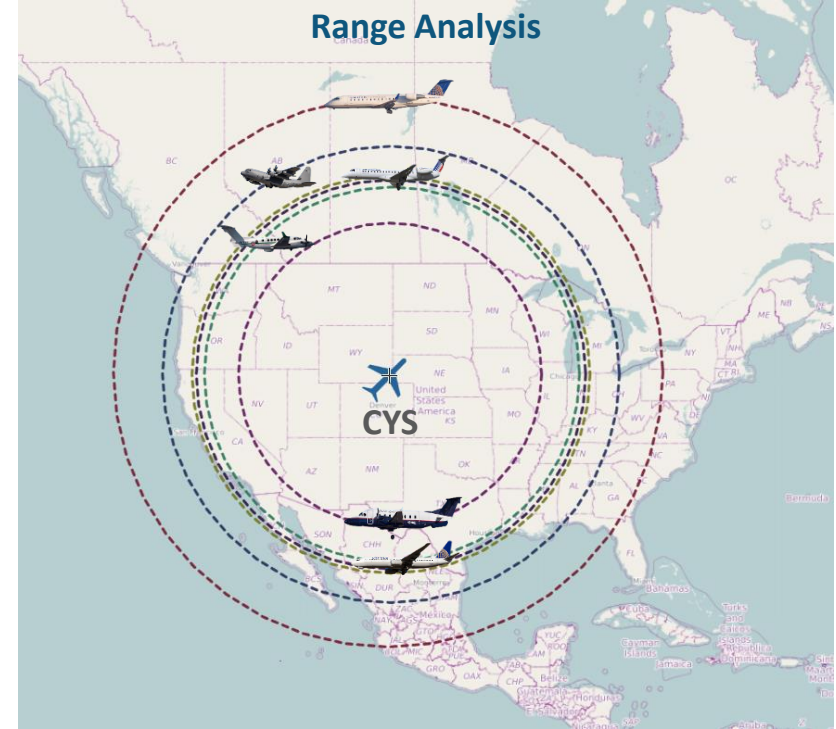
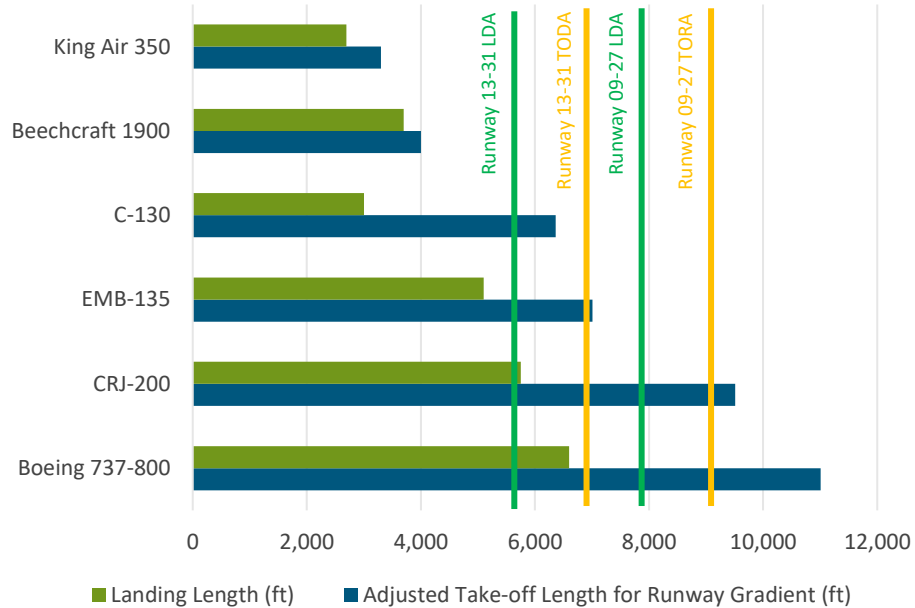
CRJ-200 (Rwy 09-27)



Boeing 737-800 (Rwy 09-27)

Runway Length & Range Analysis

Runway Length Requirements

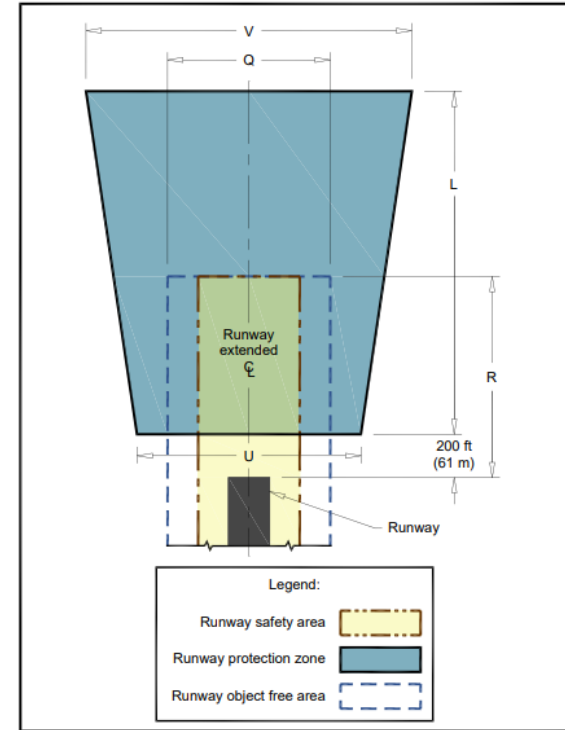


Conclusion: While additional length may be required for Runway 09-27 in the future, Runway 13-31 may instead require enhanced functionality.

Runway Design Standards

- + Evaluated Runway 09-27 and 13-31 to determine compliance with FAA AC 150/5300-13B, Change 1 Airport Design
- + Determined the following design components do not meet FAA standards
 - ROFA & RSA length beyond departure end – extends into Central Ave
 - Runway Protection Zone – In compatible development such as roads on both ends
 - Holding Position for Runway 09-27 – 12 feet to close to the runway environment
- + These nonstandard conditions will be evaluated in the alternatives development process

Figure 3-26. Runway Protection Zone (RPZ), Runway Object Free Area (ROFA), and Runway Safety Area (RSA)



Note: See [Appendix G](#) or online [Runway Design Standards Matrix Tool](#) for dimensions.

Taxiway/Taxilane Design Standards



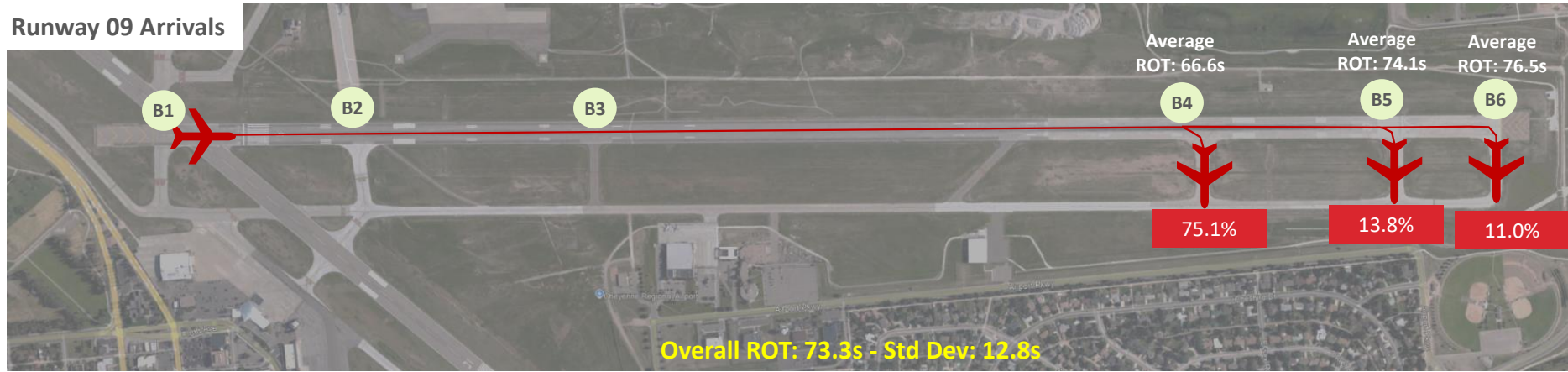
Conclusion: Nonstandard taxiway geometry will be addressed in alternatives development process

Taxiway Exit Analysis Planning Parameters

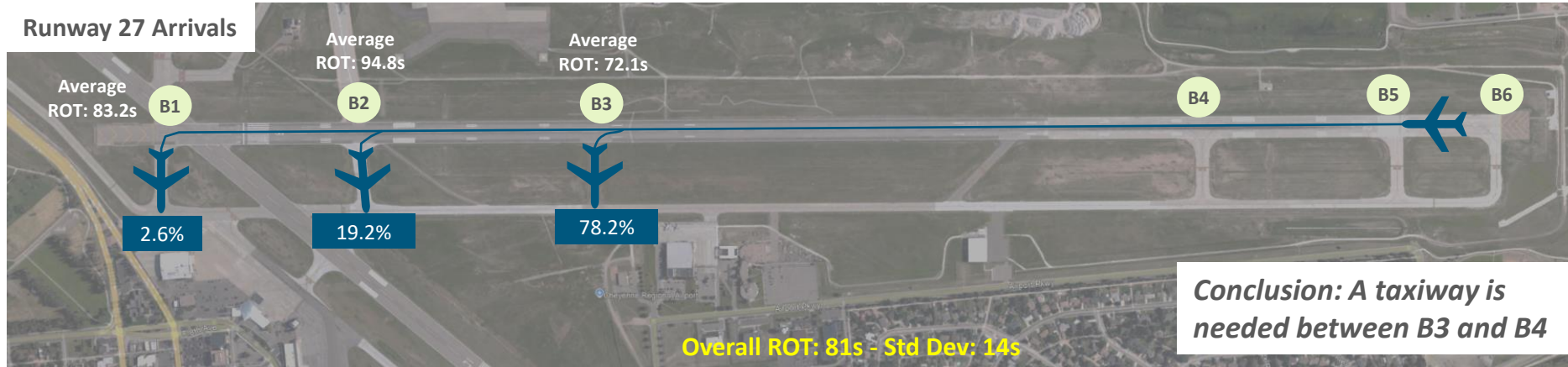
- + Evaluated the taxiway exits for Runway 09-27 to assess efficiency in aircraft movements
 - Runway Occupancy Time (ROT) is the duration an aircraft occupies the runway, measured from the moment it crosses the runway threshold (on landing) to the point where it is completely clear of the runway. FAA aims for an average ROT of 50 seconds or less for arriving aircraft.
 - Taxiway utilization is a measure to determine if a taxiway is over or underutilized. Anything above a 75% is an overutilization of a particular taxiway.
 - + Tool used- REDIM (Virginia Tech)
 - Used for planning and designing high-speed runway exits, approved by FAA.
 - Includes a built-in aircraft database covering various Aircraft Approach Categories (AAC).
 - Calculates weighted average ROT based on user-defined aircraft mix.
 - Separate evaluations were run for arrivals on Runways 9R and 27L
 - + Aircraft Fleet Mix for CYS
 - Derived primarily from existing Based Aircraft mix
 - ~4%: CRJ series (assumed for commercial operations)
 - ~13%: C-130 (assumed for military operations)
- Evaluated Runway 13-31 to determine compliance with FAA AC 150/5300-13B, Change 1 Airport Design

Runway Exit Design Interactive Model Analysis

Runway 09 Arrivals



Runway 27 Arrivals



Navigational Aids & Lighting

Navigational Aids	Airport Facility	Runway			
		09	27	13	31
Electronic Aids					
ILS OR LOC	-	-	Yes	-	-
RNAV(GPS)	-	Yes	Yes	Yes	Yes
VOR or TACAN-A	Yes	-	-	-	-
Metrological Aids					
ASOS	Yes	-	-	-	-
RVR Sensor	-	-	Yes	-	-
Visual Aids					
Edge Light System	-	HIRL	HIRL	MIRL	MIRL
Approach Lights	-	-	MALSR	-	-
Visual Slope Indicator	-	PAPI-4L	PAPI-4L	PAPI-4L	PAPI-4L
Runway End Identifier Lights	-	Yes	-	Yes	Yes
Segmented Circle with Windcone	Yes	-	-	-	-
Rotating Beacon	Yes	-	-	-	-
Supplemental Windcones	-	Yes	Yes	Yes	-

- + Instrument approach capabilities need to be improved to enhance the level of safety and flexibility for the airport
- + Per FAA AC 150/5340-30J, the Segmented Circle must include a **Traffic Pattern Indicator (TPI)**; its absence is a deficiency requiring correction



Planning Activity Level Demand

Description	Units	2024 Existing Demand	PAL 1 Likely +5	PAL 2 Likely +10	PAL 3 Likely +20	PAL 4 Optimistic + 10
Annual Traffic						
Annual Enplanements	pax	27,808	28,400	29,100	30,600	32,000
Annual Operations	ops	42,803	43,500	44,400	46,300	48,000
Departing Passenger & Operation Activity						
Departing Peak Hour Passengers	pax/hr	40	45	45	108	117
Departing Peak Hour Operations	ops	1	1	1	2	2
Arriving Passenger & Operations Activity						
Arriving Peak Hour Total Passengers	pax/hr	40	45	45	108	117
Arriving Peak Hour Operations	ops	1	1	1	2	2
Peak Hour Aircraft and Passenger Distribution						
CRJ-200 / ERJ-135	pax	40	45	45	45	45
CRJ-900	Pax	-	-	-	63	72

Terminal Requirements Methodology

- + Industry guidance utilized:
 - Airport Cooperative Research Program (ACRP), Report 25, *Airport Passenger Terminal Planning and Design*
 - International Air Transport Association (IATA) Airport Development Reference Manual (ADRM), 12th Edition
- + Space standards and wait times are measured against IATA's Level of Service (LoS) standards
- + “Optimum” LoS corresponds to stable flows, acceptable delays, and a good level of passenger comfort
- + Peak hour forecast provides demand basis for requirements to ensure facilities are sized to meet a sufficiently busy demand period

IATA Level of Service standards by functional area

Functional Area \ Level of Service Standard		SPACE STANDARDS FOR WAITING AREAS (m ² /pax)					WAITING TIME STANDARDS FOR PROCESSING FACILITIES (minutes)					PROPORTION OF SEATED OCCUPANTS (%)				
		A	B	C	D	E	A	B	C	D	E	A	B	C	D	E
		Over design	Optimum	Suboptimum	Over design	Optimum	Suboptimum	Over design	Optimum	Suboptimum	Over design	Optimum	Suboptimum	Over design	Optimum	Suboptimum
Public Departure Hall		> 2.3	2.0 - 2.3	< 2.0								> 20%	15% - 20%	< 15%		
Check-in	Self-Service Kiosk	> 1.8	1.3 - 1.8	< 1.3				< 1	1 - 2	> 2						
	Bag Drop Desk															
	All Classes	> 1.8	1.3 - 1.8	< 1.3												
	Economy Class							< 1	1 - 5	> 5						
	Business Class							< 1	1 - 3	> 3						
	Check-in Desk															
Security Checkpoint	All Classes	> 1.8	1.3 - 1.8	< 1.3												
	Economy Class							< 10	10 - 20	> 20						
	Business Class							< 3	3 - 5	> 5						
	First Class							< 1	1 - 3	> 3						
Boarding Gate Lounge	All Lanes	> 1.2	1.0 - 1.2	< 1.0												
	Regular Lane							< 5	5 - 10	> 10						
	Expedited Lane							< 1	1 - 3	> 3						
Immigration (Passport Control)	Seated Passengers	> 2.2	1.8 - 2.2	< 1.8								> 70%	50% - 70%	< 50%		
	Standing Passengers	> 1.5	1.2 - 1.5	< 1.2												
	All Lanes	> 1.2	1.0 - 1.2	< 1.0												
	Staffed Lane							< 5	5 - 10	> 10						
Baggage Claim Area	Expedited Staffed Lane							< 1	1 - 5	> 5						
	Automatic Border Control							< 1	1 - 5	> 5						
	Narrow Body	> 1.7	1.5 - 1.7	< 1.5				< 1	1 - 15	> 15						
Customs Control	Wide Body	> 1.7	1.5 - 1.7	< 1.5				< 1	1 - 25	> 25						
		> 1.8	1.3 - 1.8	< 1.3				< 1	1 - 5	> 5						
Public Arrival Hall		> 2.3	2.0 - 2.3	< 2.0								> 20%	15% - 20%	< 15%		

Source: Adapted from Airport Terminal Reference Manual (ADRM), 12th Edition, International Air Transport Association, 2022

Terminal Requirements

Description	Units	Existing Capacity	2024 Existing Demand	PAL 1 Likely +5	PAL 2 Likely +10	PAL 3 Likely +20	PAL 4 Optimistic + 10
Departures Hall							
Departures Hall	sf	~2,000	540	605	605	1,4510	1,570
Check-in							
Counters (<i>common use</i>)	pos	4	2	2	2	4	4
Counter Queue Area	sf	250	85	95	95	225	225
Airline Ticket Offices (BOH)	sf	790	350	350	350	350	350
Security Screening Checkpoint							
Screening Lanes	lanes	1	1	1	1	1	1
Queue area	sf	1,000	100	110	110	270	270
Checked Baggage Screening							
Screening Machines (EDS)	devices	1	1	1	1	1	1
Bag Screening Room Area	sf	585	600	600	600	600	600
Baggage Makeup							
Makeup Belt	lf	33	10	10	10	25	25
Makeup Area	sf	1,650	500	500	1,500	1,500	1,500
Holdrooms							
Holdroom Area	sf	3,850	1,240	1,240	2,890	2,890	2,890
Baggage Claim							
Linear Frontage	lf	52	30	35	35	80	90
Claim Devices @52 ft	devices	1	1	1	1	2	2
Arrivals Hall							
Arrivals Greeting Hall	sf	~2,000	250	250	250	580	630
Apron Parking Positions							
Apron Positions	pos	3	1	1	1	2	2

Public Parking Requirements

- + Current occupancy is ~165 spaces (based on visual survey in April 2025)
- + Escalate at annual passenger growth rate
- + Include 10% “circulation allowance” to account for mis-parked vehicles, reduce driver search time, and account for non-passenger activity, such as restaurant customers
- + Factors that could influence future requirements
 - Changes in parking price (currently free)
 - Average parker travel duration (days away from home)
 - Change in mix of local versus visiting passengers

	Base Year	PAL 1	PAL 2	PAL 3	PAL 4
Annual passengers	27,808	28,400	29,100	30,600	32,000
Public parking spaces demand	185	190	190	200	210
Public parking capacity	205	205	205	205	205
Surplus (+) Deficit (-)	+20	+15	+15	+5	-5

Employee Parking Requirements

- + Current occupancy (based on visual survey in April 2025)
 - Near terminal: ~10 spaces
 - Airport administration: ~5 spaces
- + Assume 50% higher occupancy in case survey not conducted during typical busy period
- + Escalate at blended growth of annual passengers and operations
- + Include 10% “circulation allowance” to account for mis-parked vehicles, reduce driver search time, and accommodate special events such as Board meetings
- + Existing parking requirements are adequate through the planning period, unless administrative functions are relocated to the terminal in the future

	Base Year	PAL 1	PAL 2	PAL 3	PAL 4
Annual passengers	27,808	28,400	29,100	30,600	32,000
Operations	42,803	43,500	44,400	46,300	48,000
Terminal employee spaces	15	15	15	20	20
Administration employees	10	10	10	10	10

Rental Car Requirements

- + Current occupancy is 30 spaces (based on visual survey in April 2025)
- + Assume 20% higher occupancy in case survey not conducted during typical busy period
- + Escalate at annual passenger growth rate
- + Assume service (fuel/wash/maintenance) will continue to be in remote location
- + Factors that could influence future requirements
 - Number of brands at airport
 - Extent to which rental car companies use the airport to serve the region
- + **Examine the need for a QTA facility**

	Base Year	PAL 1	PAL 2	PAL 3	PAL 4
Annual passengers	27,808	28,400	29,100	30,600	32,000
Rental car space demand	35	35	40	40	40
Ready Return Lot Capacity	40	40	40	40	40
Surplus (+) Deficit (-)	+5	+5	0	0	0

Additional Facility Requirements

Hangar and Support Facilities



Hangars

By the end of the planning horizon, provide 28,000 sq. ft. of hangar space

Combined Services Building

A designated structure for the SRE, de-icing storage, maintenance, operations and airport administration functions



Amenities & Other Service Offerings



Wash rack

Location for aircraft operators to wash their aircraft

Airport Surveillance Radar (ASR) -11

A type of surveillance equipment to enhance airspace/airfield safety



Solar farm/panels

Generate energy for the airport to help reduce operating expenses

Old Terminal Building redevelopment

A general aviation terminal that provides airport and community amenities





Land Use Analysis



Top Five Revenue Opportunities at CYS



- Full-Service FBO (Hangars + Support Services)
- Aircraft MRO for Specialized Maintenance and Paint/Interior Customization
- UAV/AAM Training, Testing & Operations
- Air Ambulance & Medical Aviation Hub



- Advanced Manufacturing Cluster for Aero Components, Specialized Assemblies, Military, Firearms, Precision Machining, etc.
- Renewable Energy which could involve Solar, SAF/Hydrogen, Electrification, Micro-Grid, or Nuclear

Top Five Revenue Opportunities at CYS



- Data Centers, Cloud Computing, AI, Big Data, and Aviation-Tech



- Specialized Logistics
- Temperature Controlled
- Paid Parking & Access Controls



- Select-Service/Extended-Stay Hotel
- Adaptive Reuse Of 1960s Terminal with Boutique Hotel, Destination Restaurant, and Museum/Events
- Small-Format Retail/F & B Pods for Convenience, Advertising, Displays, and Special Event Pop-Ups



Next Steps



Next Steps

+ Technical Work

- Finalize Preferred Development Plan
- Environmental Considerations
- Financial Analysis & Implementation Plan
- Airport Layout Plan

+ Stakeholder Engagement

- TAC Meeting No.4 – End of January
- Public Open House No.2 – End of February

+ Project Completion: March 2026

